

KL Divergence

Math Foundations — Concept 38

First, an example

Take two Bernoulli distributions on $\{0, 1\}$: a fair coin $p = (0.5, 0.5)$ and a biased coin $q = (0.9, 0.1)$. Then

$$D_{\text{KL}}(p \parallel q) = 0.5 \log \frac{0.5}{0.9} + 0.5 \log \frac{0.5}{0.1} \approx -0.294 + 0.805 \approx 0.51 \text{ nats.}$$

Read aloud: “KL of p from q equals point-five log point-five over point-nine plus point-five log point-five over point-one, approximately point-five-one nats.” Reversing the arguments,

$$D_{\text{KL}}(q \parallel p) = 0.9 \log \frac{0.9}{0.5} + 0.1 \log \frac{0.1}{0.5} \approx 0.529 - 0.161 \approx 0.37 \text{ nats.}$$

Read aloud: “KL of q from p equals point-nine log point-nine over point-five plus point-one log point-one over point-five, approximately point-three-seven nats.” Note $D_{\text{KL}}(p \parallel q) \neq D_{\text{KL}}(q \parallel p)$ — KL is asymmetric, hence *not* a metric.

1 Definition

Let p and q be probability distributions on a common measurable space $(\mathcal{X}, \mathcal{F})$, with p absolutely continuous with respect to q (written $p \ll q$). The *Kullback–Leibler divergence* of p from q is

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p \left[\log \frac{dp}{dq} \right].$$

Read aloud: “D-K-L of p given q equals the expectation under p of log of $d-p$ by $d-q$.”

Definition 1 (Discrete KL divergence). *For probability mass functions p, q on a countable set $\mathcal{X}$,*

$$D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)},$$

Read aloud: “D-K-L of p given q equals the sum over x in script- $\mathcal{X}$ of p of x times log of p of x over q of x .” with conventions $0 \log 0 = 0$ and $0 \log(0/0) = 0$. The sum is $+\infty$ if there exists x with $p(x) > 0$ and $q(x) = 0$.

Definition 2 (Continuous KL divergence). *For probability densities f_p, f_q on $\mathbb{R}^d$ (with respect to Lebesgue measure),*

$$D_{\text{KL}}(p \parallel q) = \int_{\mathbb{R}^d} f_p(x) \log \frac{f_p(x)}{f_q(x)} dx,$$

Read aloud: “D-K-L of p given q equals the integral over $\mathbb{R}$ -to-the- d of f -sub- p of x times log of f -sub- p of x over f -sub- q of x , $d-x$.” defined whenever the integrand is integrable and $\{f_q = 0\} \subseteq \{f_p = 0\}$ up to null sets.

Remark 3. D_{KL} is not symmetric and does not satisfy the triangle inequality, so it is not a metric. Nonetheless it is non-negative, vanishes only when $p = q$, and behaves additively under independence, justifying its role as a “relative information” measure.

2 Gibbs' inequality

Theorem 4 (Non-negativity of KL). *For any pair of probability distributions p, q on the same space with $p \ll q$,*

$$D_{\text{KL}}(p \parallel q) \geq 0,$$

Read aloud: “D-K-L of p given q is greater than or equal to zero.” with equality if and only if $p = q$ almost everywhere.

Proof. Write $r(x) = q(x)/p(x)$ on the support of p . By definition,

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] = \mathbb{E}_p [-\log r(X)].$$

Read aloud: “D-K-L of p given q equals the expectation under p of log of p of X over q of X , which equals the expectation under p of negative log r of X .” The function $\varphi(t) = -\log t$ is strictly convex on $(0, \infty)$. By Jensen's inequality applied to φ and the random variable $r(X)$ under p ,

$$\mathbb{E}_p [-\log r(X)] \geq -\log \mathbb{E}_p [r(X)].$$

Read aloud: “The expectation under p of negative log r of X is greater than or equal to negative log of the expectation under p of r of X .” We compute the right-hand side:

$$\mathbb{E}_p [r(X)] = \int r(x) p(x) dx = \int \frac{q(x)}{p(x)} p(x) dx = \int q(x) dx = 1,$$

Read aloud: “The expectation under p of r of X equals the integral of r of x times p of x d- x , which equals the integral of q of x over p of x times p of x d- x , which equals the integral of q of x d- x , which equals one.” since q is a probability distribution. Therefore

$$D_{\text{KL}}(p \parallel q) \geq -\log 1 = 0.$$

Read aloud: “D-K-L of p given q is greater than or equal to negative log one, which equals zero.” Because φ is *strictly* convex, Jensen's inequality is an equality iff $r(X)$ is p -almost-surely constant. Combined with $\mathbb{E}_p [r(X)] = 1$, this forces $r \equiv 1$ on the support of p , i.e. $p = q$ a.e. The discrete case is identical with integrals replaced by sums. $\square$

Remark 5. *Run with $\varphi = -\log$ replaced by an arbitrary convex φ with $\varphi(1) = 0$, the same proof yields the family of f -divergences (Csiszár). KL corresponds to $\varphi(t) = t \log t$ (after a sign convention); other choices recover total variation, Hellinger, and χ^2 divergences.*

3 Consequences

Lemma 6 (Cross-entropy decomposition). *For all p, q with $p \ll q$,*

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q),$$

Read aloud: “ H of p comma q equals H of p plus D-K-L of p given q .” where $H(p) = -\mathbb{E}_p [\log p]$ and $H(p, q) = -\mathbb{E}_p [\log q]$.

By Theorem 4, $H(p, q) \geq H(p)$ with equality iff $q = p$. Hence minimizing cross-entropy in q for fixed p is the same as minimizing KL divergence — the foundational identity behind maximum likelihood estimation and the categorical cross-entropy loss.